

usdGeospatial build loop

Functional requirements, runtime behavior and measured evidence

Implementation checkpoint

Checkpoint

35 tests passed, 0 failed, 0 skipped in this delivered run.

29 functional requirements, 9 numbered open questions and 7 grouped build stops are tracked.

The requirements and runtime derivation remain drafts. Complete scene resolution and independent OpenUSD/OV agreement remain unimplemented.

Runnable components

A read-only binding inspector reads composed USD relationships, nested overrides and forwarded targets.

A PROJ adapter converts explicit WGS 84 geographic-3D and geocentric coordinates, with operation provenance and whole-batch failures.

Input-change checks invalidate stale derivations before old tests can count as current evidence.

Anchor decoding, placement matrices, scene-target selection, general datum/grid/epoch operations and OV integration remain pending.

Coordinate checks

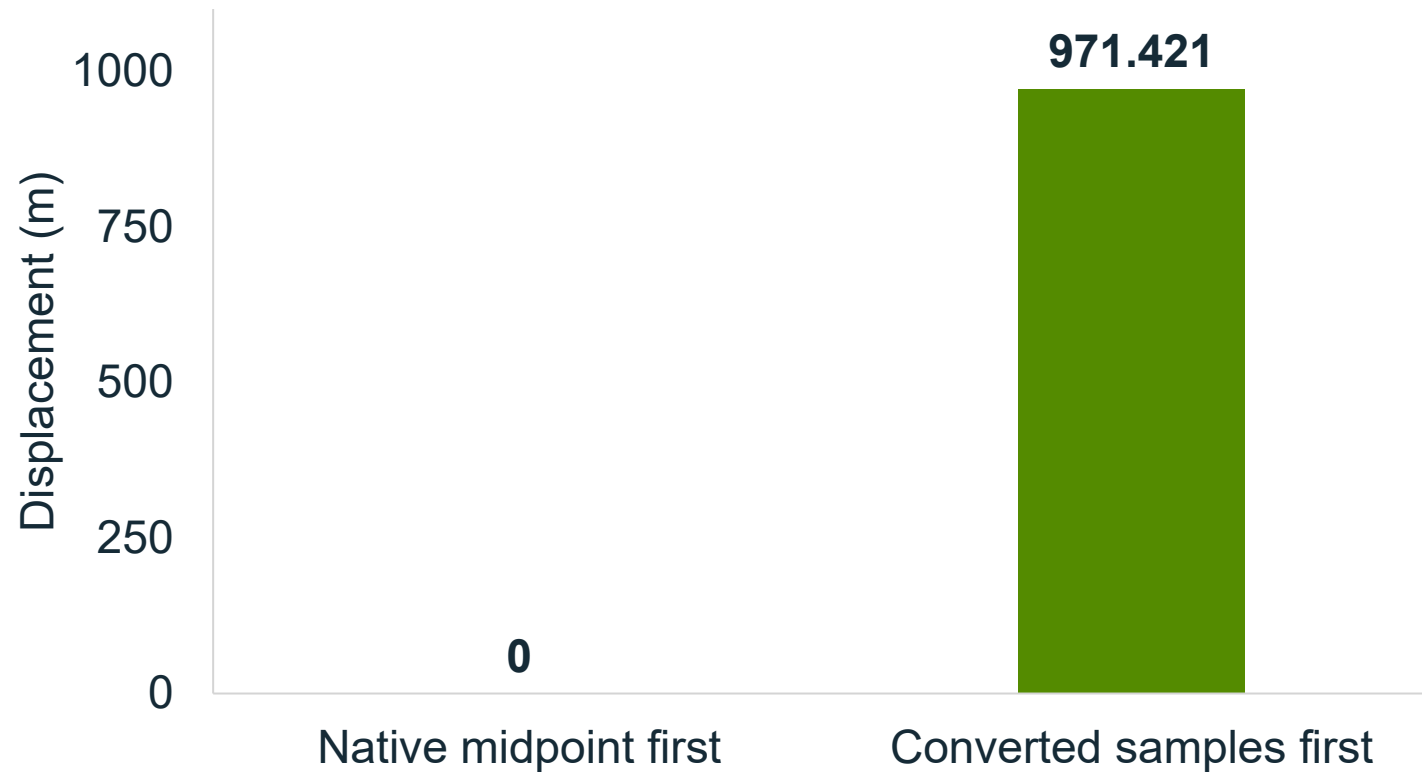
Maximum measured Cartesian residual: 0.000000000 m at a coordinate magnitude of 6378147.000 m.

Expected coordinates come from WGS 84 ellipsoid equations calculated separately from PROJ.

The analytic check uses a 0.000001 m numerical tolerance. This is a component-test tolerance.

PROJ reports this operation's accuracy as unknown. A measured residual does not replace the engine's accuracy estimate.

Sampled position path



This exercises requirement 20, Positions between recorded moments, using its equatorial example.

Whether USD may record geographic positions, and which carrier holds them, remains open.

Partner baseline

Tower-origin residual: 0.000000 m against the partner's stated coordinates.

Tower height: 299.999995 m against the stated approximately 300 m.

Calibrated grid-origin residual: 0.001339 mm.

These checks reproduce the existing stock-USD example. They do not validate a new scene resolver.

The attachment remains local because redistribution has not been established. Public reruns skip its two checks.

Open decisions

S01: position carrier, geographic recording and the position/offset boundary.

S02: preserving an asset's native CRS while placing it in a project.

S03: scene units/up axis, the placement basis and its distance/extent bound.

S04: consumer target selection and geographic scene output.

S05: dependency declaration, explicit export and re-resolution.

S06: practitioner controls, operation resources, a second engine and OV evidence.

S07: complete rules for hierarchical binding relationships.

Delivery for each run

A successful test invocation produces an immutable checkpoint with the input revision, source hashes, environment, results and remaining decisions.

Run evidence updates the README. The PR body and slide content are then derived from the README.

A publication check rejects issue/PR citations, private paths and private communication links before a push or PR update.

One standing draft PR receives each reviewed checkpoint. Earlier run records remain in the branch history.